Post-Ontological Observer Engineering: Compiling Self-Referential Quantum Observers into the PORE Protocol for Collapse, Agreement, and Control

0) Reader contract and claim level

This paper is written as an **operational synthesis**, not a metaphysical manifesto. It merges (i) a formal theory of **self-referential observers** in quantum dynamics—where an observer is an adaptive process with a trace, filtration, measurable policy, and Born-rule kernels—and (ii) **PORE/PoE**, a protocol-first framework that compiles loop-bearing dynamics into reproducible effective coordinates under explicit intervention channels and falsifiability gates.

To avoid category errors, we state at the outset what counts as a contribution and what does not.

0.1 What is claimed (and in what sense)

Claim C1 (operational equivalence, not ontological identity).
We claim a disciplined equivalence between two descriptions:

- the observer-side description: adaptive kernel generation and selective state update along a trace, and
- the PORE-side description: protocol-fixed logging and compilation into effective coordinates validated by harness gates.

This equivalence is formalized as an **Observer→PORE compiler**: a mapping from observer traces/logs to protocol-bound artifacts ($\Xi\hat{}$, gates, and optional gain estimates) that remain stable only where the harness passes.

Claim C2 (reproducible protocol artifacts).
We claim that the merged framework produces *portable, checkable objects* that a reader can implement:

- a protocol specification $P=(B,\Delta,h,u)$,
- a one-page compiler card (Appendix A),
- gate tests (proxy stability and probe backreaction),
- and a minimal experiment protocol (MEP) for estimating local operator gains.

These artifacts are the primary "deliverables" of the paper. Their job is to prevent silent shifts in boundary, measurement, and meaning—what we call **ontology drift**—by forcing any claim to be indexed to a declared protocol and validated by explicit acceptance/rejection rules.

Claim C3 (collapse/agreement/objectivity are operationally demoted, but mathematically sharpened).
Within this merged stance:

- "collapse" is treated as **internal conditional certainty** relative to a filtration (plus policy-induced latching),

- "agreement" is treated as a theorem under **frame map + compatibility + record accessibility**, and
- "objectivity" is treated as a redundancy phenomenon (SBS-style encodings) supporting high-probability consensus.

No new physical postulates are added; the gain is a clearer separation between what is tautologically true (conditional certainty), what is conditionally derivable (agreement/objectivity), and what must be validated empirically (protocol stability and probe identifiability).

0.2 What is not claimed (explicit non-goals)

Non-claim N1 (not a new interpretation of QM, not a new collapse law).
We do not propose a novel dynamical collapse mechanism or an ontological reading of the quantum state. The observer model is built from standard instrument calculus and measurability conditions; its "collapse" result is internal and conditional, not a new law of nature.

Non-claim N2 (not a global Theory of Everything).
Although the PORE/PoE language is intentionally general, the paper does not claim a complete unification of all physics or a final ontology. "Universality" here is methodological: a portable protocol grammar plus falsifiability harness that can be applied to many loop-bearing systems.

Non-claim N3 (not a unique microphysical decomposition).
We do not claim that every phenomenon admits a unique "true" decomposition into hidden degrees of freedom. The framework is explicitly **boundary-relative**: different choices of B,Δ,h,u can yield different effective objects and different compiled coordinates, and comparability must be demonstrated rather than assumed.

0.3 How to read the rest of the paper

If you read this paper as "what exists," it will disappoint you. If you read it as "what can be stably described, compared, and controlled under declared protocols," it is meant to be immediately usable.

- Sections 2–3 fix the protocol object and the observer backbone.
- Sections 4–6 derive internal collapse, agreement/objectivity conditions, and an invariant geometry for comparing probe changes.
- Sections 7–9 provide the compiler proposition, harness gates, and a minimal gain-estimation protocol.

Everything is written so that a reader can reproduce the artifacts and rerun the acceptance criteria. That is the contract.

1) Motivation: why "observer inside dynamics" needs a harness

The quantum measurement problem is often described as a tension between two perfectly usable rule sets—unitary evolution for closed systems and stochastic updates associated with measurement—while the ontological status of "collapse" remains unsettled across interpretations.
This paper's motivation is to keep what is technically sharp (the operational formalism) while removing two recurring sources of conceptual drift: **(i)** treating "the observer" as an external narrator rather than a definable process inside the dynamics, and **(ii)** arguing about "what is

real" without a protocol-level falsifiability harness that prevents silent changes of boundary, probing, or meaning.

1.1 Two missing ingredients in many collapse debates

Missing ingredient A: an internal observer model (trace + policy + kernels)

A large fraction of collapse talk lacks a mathematically explicit model of an observer that (a) records outcomes, (b) chooses future measurements based on those recorded outcomes, and (c) therefore makes "measurement" part of the system's causal loop. The self-referential observer framework addresses this directly by modeling an observer as an adaptive stochastic process internal to standard quantum theory—trace, measurable policy, CP instruments, Born-rule kernels, and between-tick CPTP evolution—proved to induce a unique law on infinite outcome histories (via Ionescu–Tulcea) under mild regularity.

Crucially, this yields two internal phenomena "for free" (no extra postulates):

- **delta-certainty**: past outcomes become certain within the observer's own filtration;
- **latching**: history-dependent policies make counterfactual branches diverge irreversibly, reproducing the phenomenology that collapse debates try to explain.

Missing ingredient B: engineering gates that stop "ontology drift"

Even with a correct internal observer model, discussions still drift if we allow unstated changes in *what is being measured*, *what is treated as inside the system*, or *what interventions are permitted*. PORE's central move is protocol discipline:

(1.1) $\mathbf{P = (B, \Delta, h, u)}$, with $u \in \{\text{Pump, Probe, Switch, Couple}\}$.

It then insists that any "effective object" and any compiled coordinate only exist **under** that declared protocol, and are accepted only if they pass falsifiability gates—especially proxy stability (Gate 1) and probe backreaction (Gate 3).
This is explicitly presented as an "anti-handwaving adaptor": if a gate fails, you do not patch narratives—you route to protocol repair (boundary, probing, compilation, regime segmentation).

1.2 Aim: replace "interpretation talk" with a two-part operational program

The merged aim is to translate foundational questions into a reproducible workflow:

(a) Well-posed observer stochastic law (internal collapse, agreement conditions).
We treat observers as adaptive stochastic processes defined by measurable kernels and filtrations, so "collapse" becomes an internal conditional-certainty fact and "agreement" becomes a theorem under explicit conditions (frame map + compatibility + record accessibility), with objectivity emerging from redundancy (SBS).

(b) Protocol-fixed operational compilation + falsifiability harness (PORE).
We treat any effective claim as a compiled artifact from logged trajectories under a declared protocol, guarded by gates that detect instability and observer backreaction—particularly the discipline that "Probe must not secretly be Pump/Switch/Couple."

1.3 What changes (conceptually) once you have both

With (a) and (b) in place, long-standing ambiguities are reframed as testable questions:

- "Is collapse real?" becomes: **Is the observer process well-posed, and what filtration-fixedness/latching does it imply?**
- "Do observers agree?" becomes: **Do we have a frame map, compatible effects (or a joint measurement), and an accessible record?**
- "Is the object stable?" becomes: **Under fixed P, do compiled coordinates remain stable (Gate 1) and does probing remain identifiable (Gate 3)?**

That is the motivation for the rest of the paper: build the observer backbone (Sections 3–6), then compile it into protocol-fixed coordinates with explicit failure routing (Sections 7–9), so philosophical disagreement is systematically converted into an engineering-grade empirical question.

2) PORE / PoE protocol object (protocol-first stance)

PORE begins by **refusing to start from ontology** ("what the system really is") and instead fixes a **protocol** that determines what counts as data, what counts as an intervention, and what counts as an admissible claim. The central move is: *if you cannot state your protocol, you cannot state your object*—because any "object" in this framework is an **observable, loop-bearing effective regularity under a declared interface**, not a free-floating metaphysical entity.

2.1 The protocol package

We define a protocol as the following tuple:

(2.1) $\mathbf{P = (B, \Delta, h, u)}$

where:

- **B (boundary)**: a *declaration* of what is treated as "inside" the loop-object versus "outside / exogenous," including what is held fixed (or treated as noise) during compilation and testing.
- **Δ (timebase / sampling step)**: the observation cadence that defines what one "tick" means for logging, estimation, and control.
- **h (observation map)**: the measurement/compression operator that turns the (possibly high-dimensional) internal microstate into a logged macro-observable.
- **u (admissible operator channels)**: the *named intervention knobs* allowed under the protocol, treated as first-class and reportable.

This is the **anti-ontology-drift mechanism**: if you later change what you consider "inside" (B), change the cadence (Δ), change the sensor/compression (h), or change what you're allowed to do (u), then you have not "refined the same model"—you have **changed the protocol**, and comparability must be re-earned rather than assumed.

2.2 Logging: what the compiler is actually allowed to see

Under a fixed protocol, the compiler does not see the world "as it is," but only the protocol-bound log:

(2.2) $\mathbf{z[n] = h(x(t_0 + n\Delta))}$

Everything downstream (loop detection, coordinate compilation, gain estimation, falsification

gates) must be computed from **z[n]** (or declared embeddings of it), under the fixed P. This is not a limitation; it is the discipline that prevents silently importing "extra knowledge" when a story becomes convenient.

2.3 Minimal operator grammar: the admissible channels

PORE standardizes a minimal control/interaction vocabulary:

(2.3) **u ∈ {Pump, Probe, Switch, Couple}**

At this stage we only need the **protocol meaning** (the full operational signatures come later):

- **Pump**: interventions that reshape the loop's "resource / basin" structure (stabilize, deepen, energize, replenish).
- **Probe**: interventions that change what is queried/observed, explicitly admitting potential backreaction and requiring it be tested rather than ignored.
- **Switch**: interventions (or endogenous events) that trigger regime changes; treated as reportable objects rather than "noise" to be patched away.
- **Couple**: interventions that increase closure/binding/confinement and reduce leakage (often the mechanism behind robust "objectivity" at the effective level).

2.4 Loop validity prerequisite: "Ξ exists only if loop exists"

The protocol-first stance makes a hard, falsifiable claim:

> **Effective coordinates Ξ are only defined for loop-bearing objects under a fixed protocol.**

Operationally, PORE treats "objecthood" as: **recurrence + stable closure law + bounded leakage** under P.
A concise statement of the prerequisite is:

(2.4) **Loop valid ⇔ recurrence ∧ return-map stability ∧ bounded leakage**

And the philosophical consequence is explicit:

(2.5) **If Gateo fails, Ξ is undefined.**

Why this matters philosophically (without metaphysics)

- **No "object" ⇒ no coordinate.** If your declared protocol cannot exhibit a stable recurrent closure (or leaks too fast / drifts / has no stable return law), then any attempt to report a low-dimensional "state of the object" is *not wrong*—it is **ill-posed**.
- **Failure produces actions, not narratives.** When Ξ is undefined, the response is not an interpretive patch ("the system is mysterious"), but a protocol repair: revise **B**, redesign **h**, change **Δ**, or split the system into multiple loops.

This is the core "post-ontological" discipline: **claims are only made inside the space where the protocol makes them identifiable and stable**—and outside that space, the framework forces you to say "undefined," not "true but hard to measure."

3) Self-referential observer skeleton (quantum-side hard backbone)

This section states the **minimal observer model** used throughout: an observer is not an "external narrator," but an **adaptive stochastic process** whose future measurement choices depend measurably on its recorded past. The construction is deliberately standard: outcomes come from the Born rule applied to a quantum instrument, and the post-measurement state updates by the corresponding selective CP map.

3.1 Outcome alphabet, contexts, trace, filtration

We fix:

- **Outcome alphabet** Φ, assumed finite or countable (discrete σ-algebra).
- **Context (channel) space** Θ, assumed standard Borel (e.g., Polish with its Borel σ-algebra).

Let $\varphi_k \in \Phi$ denote the realized outcome at tick k, with prefix:

(3.1) $\boldsymbol{\varphi_{1:k} := (\varphi_1,\ldots,\varphi_k) \in \Phi^k}$

The observer's "memory" is formalized as the filtration generated by its trace:

(3.2) $\boldsymbol{\mathcal{F}_k := \sigma(\varphi_1,\ldots,\varphi_k)}$ (with $\mathcal{F}_0$ trivial).

The pair $(\varphi_k, \mathcal{F}_k)$ is the minimal "internal observer" structure: **what happened** and **what is knowable at tick k**.

3.2 Adaptive policy (self-referential context selection)

An observer is *self-referential* because it selects the next measurement setting as a measurable function of its own past outcomes:

(3.3) $\boldsymbol{f_k: \Phi^{k-1} \to \Theta}$ is Borel-measurable (adaptive policy).

(3.4) $\boldsymbol{\theta_k := f_k(\varphi_{1:k-1})}$ (chosen context at tick k).

A key regularity assumption ensures the resulting kernels are well-defined as measurable objects (needed later for existence/uniqueness on Φ^∞):

(3.5) $\boldsymbol{(\theta,\sigma) \mapsto \mathrm{Tr}[M_{\theta,\varphi}(\sigma)]}$ **is jointly measurable** (e.g., via continuity of $\theta \mapsto M_{\theta,\varphi}$ in a strong/diamond norm sufficient for measurability).

3.3 Instruments and effects (Born-rule-ready interface)

Let $D(H)$ be the density operators on a fixed Hilbert space H (the "world" degrees of freedom relevant at the chosen modeling level). A **quantum instrument** is a family of completely positive, trace–nonincreasing maps indexed by context θ and outcome φ:

(3.6) $\boldsymbol{M_{\theta,\varphi}: D(H) \to D(H)}$ is CP and trace–nonincreasing.

Instrument normalization (trace preservation after averaging over outcomes):

(3.7) $\boldsymbol{\sum_{\varphi\in\Phi} M_{\theta,\varphi}}$ **is CPTP** for each θ.

Define the associated **effect operators** using the dual (Heisenberg) map M*:

(3.8) **$E_{\theta,\varphi} := M^*_{\theta,\varphi}(I)$**, with $0 \le E_{\theta,\varphi} \le I$ and $\Sigma_\varphi E_{\theta,\varphi} = I$.

This makes probabilities legible both ways:

(3.9) **$Pr(\varphi \mid \theta, \sigma) = Tr[M_{\theta,\varphi}(\sigma)] = Tr[E_{\theta,\varphi}\, \sigma]$**.

(Effects $E_{\theta,\varphi}$ will later be the objects on which "compatibility/commutation" is imposed when discussing cross-observer agreement.)

3.4 Tick dynamics (latent evolution → sampling kernel → selective update)

The observer operates in discrete ticks k = 1,2,… with **latent between-tick dynamics** and **measurement only at ticks**. Between ticks the state evolves by a CPTP map:

(3.10) **$\rho_{k^-} := E_k(\rho_{k-1})$**, where each E_k is CPTP.

Given the realized history $\varphi_{1:k-1}$ (hence θ_k), the model induces a Born-rule **transition kernel** on Φ:

(3.11) **$K_k(\varphi_{1:k-1}, \varphi) := Tr[M_{f_k(\varphi_{1:k-1}),\varphi}(\rho_{k^-})]$**.

Outcome sampling at tick k is then:

(3.12) **$\varphi_k \sim K_k(\varphi_{1:k-1}, \cdot)$**.

Finally, the **selective post-measurement update** is:

(3.13) **$\rho_k := M_{\theta_k,\varphi_k}(\rho_{k^-}) / Tr[M_{\theta_k,\varphi_k}(\rho_{k^-})]$**.

Two immediate sanity properties are built in by normalization:

- Kernel normalization:

(3.14) **$\Sigma_{\varphi\in\Phi} K_k(\varphi_{1:k-1}, \varphi) = 1$**.

- Realized outcomes are almost surely nonzero-probability, so the conditional update is well-defined on-path.

3.5 Why this skeleton is the "hard backbone" for the merge

This observer model is intentionally *minimal but rigid*:

- Minimal: it assumes only standard instrument calculus + measurable adaptive policies + CPTP latent evolution.
- Rigid: the moment measurability/regularity fails, the global law on Φ^∞ can fail to exist (so "observer" is not hand-waving, but a mathematically delimited construct).

In the next section we use these ingredients to construct the **unique global stochastic law** on infinite traces and extract the two key internal phenomena—**delta-certainty** and **latching**—that will later align with PORE's protocol-first gates (especially probe backreaction discipline and regime-change detection).

4) Existence/uniqueness ⇒ "collapse" as internal conditional certainty

Section 3 defines an adaptive observer by a family of **history-indexed kernels** {K_k} on outcomes. Section 4 extracts the key consequence: under mild measurability conditions, the observer's infinite trace is a **well-posed probability law** on Φ^∞, and "collapse" appears as a strictly internal statement about conditional certainty relative to the observer's own filtration—together with an irreversible branching mechanism when the observer's policy depends on its past.

4.1 From local kernels to a unique global law on Φ^∞

Let (Φ, Σ_Φ) be the measurable outcome space (for discrete Φ, Σ_Φ is the power set). For each tick k, Section 3 defined a measurable transition kernel

(4.0) **K_k: (Φ^{k−1}, Σ_Φ) → [0,1]**, K_k(φ_{1:k−1}, ·) a probability measure.

Assume the standard regularity:

- (A1) **Measurability:** for every measurable A ⊆ Φ, the map φ_{1:k−1} ↦ K_k(φ_{1:k−1}, A) is Σ_{Φ^{k−1}}-measurable.

This assumption is ensured by the measurability of the adaptive policy f_k and the joint measurability of (θ,σ) ↦ Tr[M_{θ,φ}(σ)] (Section 3).

Then the Ionescu–Tulcea extension theorem applies: the family {K_k} defines a **unique probability measure** on the infinite product space Φ^∞ consistent with the finite-dimensional cylinders.

(4.1) **∃! P on (Φ^∞, Σ_{Φ^∞}) such that**
for all k and all measurable A₁,…,A_k ⊆ Φ,
$$\mathbf{P(\varphi_1\in A_1,\dots,\varphi_k\in A_k) = \int_{A_1} K_1(d\varphi_1) \int_{A_2} K_2(\varphi_1,d\varphi_2) \dots \int_{A_k} K_k(\varphi_{1:k-1},d\varphi_k).}$$

This is the mathematical "closure" of the observer: *once you can specify the kernels measurably, the observer's entire life-history is a single well-defined stochastic object.*

4.2 Internal delta-certainty (collapse as filtration-fixedness)

Recall the observer filtration F_k := σ(φ₁,…,φ_k). Under the global law P, each φ_j (j≤k) is F_k-measurable. Therefore the conditional distribution of φ_j given F_k is almost surely a point mass at its realized value. In discrete Φ this can be written as:

(4.2) **P(φ_j = x | F_k) = 1{ x = φ_j }**, for j ≤ k.

Equivalently: once the observer has recorded outcomes up to time k, the past outcomes are **delta-certain relative to the observer's own information algebra**. This is the sense in which the framework formalizes "collapse" without inserting any new physical postulate: the "collapsed past" is simply the tautology that realized random variables become certain when conditioning on the σ-algebra that contains them.

Operational reading: collapse is not "the world changed," but "the observer's internal bookkeeping has fixed a value inside F_k."

4.3 Latching irreversibility from history-dependent policies

Delta-certainty alone is not yet "branching"; branching enters when the observer's future kernels depend on its past outcomes through the adaptive policy:

(4.3) **$\theta_{k+1} = f_{k+1}(\varphi_{1:k})$**.

If f_{k+1} is non-constant on a set of histories with nonzero probability, then different realized histories imply different future contexts, hence different instruments, hence different future kernels. Concretely, there exist histories $a,b \in \Phi^k$ with $P(\varphi_{1:k}=a)>0$ and $P(\varphi_{1:k}=b)>0$ such that:

(4.4) **$f_{k+1}(a) \neq f_{k+1}(b) \Rightarrow K_{k+1}(a,\cdot) \neq K_{k+1}(b,\cdot)$** (generically).

This is **latching**: once an outcome has occurred, the observer's own policy can "lock in" a future measurement regime, making different branches not merely epistemically different, but *dynamically different* in what they will probe next.

Irreversibility is then structural: even if two histories later converge to the same macro-summary under some coarse h, their induced future kernels can remain different because the policy remembers details in $\varphi_{1:k}$. In other words, the "arrow" is implemented as **path dependence of the observation/control interface** rather than as an extra thermodynamic axiom.

4.4 Interpretive punchline (kept operational)

Putting 4.2 and 4.4 together yields the merged-paper definition of collapse compatible with PORE's protocol-first stance:

- **Collapse = filtration-fixedness:** realized events become delta-certain when conditioned on the observer's own record algebra F_k.
- **Branching = policy-induced latching:** the observer's adaptive choice of future contexts makes branches generatively distinct via different kernels.

No extra collapse postulate is required. What remains "physical" (and will matter in the PORE merge) is not the logical fact of conditional certainty, but the **operational costs and signatures** of branching: probe backreaction, record accessibility, compatibility constraints, and whether a given protocol can compile stable effective coordinates across branches.

5) Cross-observer agreement and objectivity (records + compatibility + redundancy)

Section 4 established "collapse" as **internal** (filtration-fixedness). We now ask a harder question: when do *two different observers* (with different traces, instruments, and possibly different contextual parametrizations) obtain **the same delta-certainty** about "the same event"? The answer is **AB-fixedness**: agreement becomes a theorem once three operational requirements are met—(i) a frame map aligning propositions, (ii) compatibility ensuring a joint probability model exists, and (iii) an accessible record so conditioning is meaningful.

5.1 AB-fixedness: agreement as a conditional-certainty theorem

5.1.1 Frame map: "same proposition" must be well-defined

Let observer A use contexts Θ^A and outcomes Φ, and observer B use contexts Θ^B and (possibly relabeled) outcomes Φ. A **frame map** from A to B is a triple:

(5.1) **T := (T_Θ, T_Φ, T_E)**

with:

- **T_Θ: Θ^A → Θ^B** measurable (context alignment),
- **T_Φ: Φ → Φ** measurable (outcome relabeling),
- **T_E** structure-preserving on effects, satisfying:

(5.2) **T_E(E^A_{θ,φ}) = E^B_{T_Θ(θ), T_Φ(φ)}**.

Operational meaning: without T, the phrase "B agrees with A" is ill-posed, because A's and B's propositions are not yet the *same* proposition.

5.1.2 Compatibility: joint probability must exist

Agreement is not even definable if A's and B's statements cannot be embedded into a single joint model. The sufficient condition is **commutation** (or, more generally, **joint measurability**):

(5.3) **[T_E(E^A_{θ,φ}), E^B_{θ′,φ′}] = 0**, with θ′ = T_Θ(θ), φ′ = T_Φ(φ).

Equivalently, there exists a joint POVM {F_{u,v}} such that each observer's effect is a marginal of the same joint measurement.

5.1.3 Accessible record: conditioning must have an object

A **record variable** is a random variable R (valued in Φ) that equals A's realized outcome almost surely on the event of that outcome, and is *accessible* to B at some tick k in the strict sense:

(5.4) **R = φ (a.s. on "A realized φ under θ"), and R is F_k^B-measurable.**

Records can be realized by direct trace sharing, shared classical registers, or redundant environment encodings that B can read later.

5.1.4 AB-fixedness statement (the "agreement theorem")

Under (i) frame map, (ii) compatibility, and (iii) accessible record, B's conditional probability for the mapped outcome becomes delta-certain upon conditioning on B's own filtration:

(5.5) **P^B(outcome equals T_Φ(R) | F_k^B) = 1.**

This is AB-fixedness: *cross-observer agreement is not a metaphysical axiom; it is a conditional-certainty consequence of record accessibility plus compatibility within an aligned frame.*

Policy-independence note. AB-fixedness is a statement about conditional probabilities on an already-generated σ-algebra; subsequent adaptive policy choices do not retroactively change the conditional certainty at F_k^B.

5.2 Objectivity via redundancy: SBS / "quantum Darwinism" as multi-observer upgrade

AB-fixedness is pairwise. "Objectivity" is the many-observer strengthening: many observers read

different pieces of the environment and still converge (with high probability) to the same claim about S. The formal engine is a **spectrum broadcast structure (SBS)**: the environment carries many redundant "copies" of the same classical label i.

5.2.1 SBS form (pointer decomposition)

Let S be the system and $E_1,\ldots,E_m$ be environment fragments. SBS asserts the joint state has the classical–quantum form:

(5.6) $$\rho_{S E_1 \ldots E_m} = \sum_i p_i |\psi_i\rangle\langle\psi_i|_S \otimes \left(\bigotimes_{j=1}^{m} \rho_{E_j}^{(i)}\right).$$

Interpretation: the label i is "broadcast" into multiple fragments; each fragment E_j carries a local conditional state $\rho_{E_j}^{(i)}$ that encodes i.

Define a readable set of fragments $J \subseteq \{1,\ldots,m\}$. The **redundancy level** is:

(5.7) $$\text{Redundancy} = |J|.$$

5.2.2 Compatibility as non-demolition (local readouts must not fight the pointer)

Each observer A_j reads only E_j using a local instrument whose effects commute with the pointer projections on S (or are jointly measurable with them). This "non-demolition" condition is exactly the multi-observer version of compatibility needed for AB-fixedness to be meaningful across many reports.

5.2.3 Distinguishability: readable fragments must actually separate labels

SBS becomes operational when the fragment states $\{\rho_{E_j}^{(i)}\}$ are distinguishable across i. Several equivalent criteria can be used; two common ones are:

- Trace-distance gap:

(5.8) $$\tfrac{1}{2}\left\|\rho_{E_j}^{(i)} - \rho_{E_j}^{(i')}\right\|_1 \geq \delta_j > 0, \text{ for } i \neq i'.$$

- Quantum Chernoff exponent:

(5.9) $$\xi^{(j)}(i,i') := -\log \min_{s\in[0,1]} \mathrm{Tr}\left[\left(\rho_{E_j}^{(i)}\right)^s \left(\rho_{E_j}^{(i')}\right)^{1-s}\right] > 0.$$

Either way, the point is the same: each readable fragment provides **independent evidence** for the same i.

5.2.4 High-probability consensus improves with redundancy (readable fragments)

In the binary case $i\in\{0,1\}$, define per-fragment Chernoff exponents $\xi^{(j)}$ and the aggregate exponent:

(5.10) $$\Xi_J := \sum_{j\in J} \xi^{(j)}.$$

Then the optimal collective discrimination error on $\bigotimes_{j\in J} E_j$ admits an exponential bound:

(5.11) $$P_{\text{error}}^{(J)} \leq \tfrac{1}{2} \exp(-\Xi_J).$$

If fragments are i.i.d. with $\xi^{(j)}=\xi>0$, this becomes:

(5.12) $$P_{\text{error}}^{(J)} \leq \tfrac{1}{2} \exp(-\xi|J|).$$

Hence consensus tightens exponentially with redundancy: with compatible local readouts and sufficiently distinguishable fragment encodings, the probability that two observers disagree vanishes rapidly as |J| grows (and is exact in the noiseless, exact-SBS limit).

5.3 How this plugs into PORE (one operational bridge sentence, not a new ontology)

In the merged framework, **agreement/objectivity become protocol-sensitive engineering properties**: "accessible record" must fall inside the declared boundary B and be readable via h (or via explicitly declared side-channels), while "compatibility" is the non-negotiable condition that prevents fabricating agreement by post-hoc relabeling or incompatible probes. Redundancy (SBS-style) is then the many-copy mechanism that makes "Couple" (closure/lock-in) robust enough for stable, shared $\hat{\Xi}$-coordinates to exist across observers, rather than only within a single observer's private trace.

6) Collapse-frame geometry as a protocol bridge (tick-time vs channel distance)

Section 5 treated *agreement/objectivity* as an interplay of **records + compatibility + redundancy**. Section 6 adds a geometric layer that is deliberately modest: it does **not** decide commutation (that's algebra), but it gives an **invariant classifier** for separations between measurement events that differ both in **tick index** and **channel/context choice**—so "small probe change vs regime change" remains meaningful across observer frames and protocol variants.

6.1 Channel manifold and metric (what "channel distance" means)

Let Θ be the channel/context space (Section 3), now viewed as a manifold equipped with a Riemannian metric g that encodes "operational dissimilarity" of channels (task-specific, Fisher-type, or information-geometric choices are all admissible).

For $\theta,\theta' \in \Theta$ define the (local) squared norm:

(6.0) $$||\Delta\theta||_g^2 := g(\theta-\theta', \theta-\theta')$$

with $\Delta\theta := \theta-\theta'$ in a coordinate chart (or replace by squared geodesic distance for finite separations).

6.2 Collapse interval (signed quadratic form)

A measurement event is a pair (τ,θ) where $\tau \in \mathbb{Z}$ is the discrete tick index and $\theta \in \Theta$ is the chosen context/instrument parameter. Fix a time-scale constant $T_c > 0$ that calibrates "one tick" against "one unit of channel displacement."

(6.1) $$s_c^2((\tau,\theta),(\tau',\theta')) := (T_c)^2(\Delta\tau)^2 - ||\Delta\theta||_g^2$$

with $\Delta\tau := \tau-\tau'$ and $\Delta\theta := \theta-\theta'$.

This induces a sign classification:

- (6.2a) $s_c^2 > 0$: "time-like" (tick separation dominates)
- (6.2b) $s_c^2 = 0$: "null"

- (6.2c) $\mathbf{s_c^2 < 0}$: “channel-like” (channel displacement dominates)

Protocol bridge intuition. In PORE terms, τ is the *tick-time* induced by the protocol timebase Δ, while θ represents the *probe channel* choice (what interface you coupled / what you measured). The interval is a compact way to say: “did we mostly wait within a comparable probing regime, or did we move far in probe-space?”

6.3 Collapse-frame isometries G_c (the geometry-preserving maps)

Define the collapse-frame isometry group as the transformations preserving s_c^2 for all pairs of events:

(6.3) $\mathbf{G_c := \{\, T: \mathbb{Z}\times\Theta \to \mathbb{Z}\times\Theta \text{ bijective} \mid s_c^2(e_1,e_2)=s_c^2(T(e_1),T(e_2)) \ \forall e_1,e_2 \,\}}$

Canonical subgroups (the robust ones in discrete τ):

- Time translations: $(\tau,\theta) \mapsto (\tau+n, \theta)$
- Channel isometries: $(\tau,\theta) \mapsto (\tau, \phi(\theta))$ with ϕ an isometry of (Θ,g)

Mixed “boost-like” transforms can exist in special designs (e.g., Euclidean Θ with constant metric and a rational lattice compatible with $\tau \in \mathbb{Z}$), but discreteness of τ makes them optional and model-dependent; the safe default is block-diagonal (time translations × channel isometries).

6.4 The warning: geometry does not act on effects (you need *-automorphisms)

The interval only talks about labels (τ,θ). But agreement/incompatibility ultimately depends on the operator algebra of effects $E_\{\theta,\varphi\}$. Hence a geometric map T must be paired with an effect-level structure map T_E that transports the measurement propositions themselves:

(6.4) $\mathbf{T_E(E^A_\{\theta,\varphi\}) = E^B_\{\theta',\varphi'\}}$ whenever $(\tau',\theta')=T(\tau,\theta)$ and $\varphi'=T_\Phi(\varphi)$.

Assumption C in the observer paper requires T_E to arise from a *-automorphism on the relevant world algebra factor (e.g., unitary/antiunitary conjugation), so commutators are preserved:

(6.5) $\mathbf{[T_E(X), T_E(Y)] = T_E([X,Y]).}$

This is the non-negotiable caution: **interval invariance is not, by itself, algebraic.** Without the accompanying T_E, a purely geometric relabeling can *fabricate* apparent compatibility (or destroy real compatibility) by distorting channel distances while leaving the underlying operators unchanged.

6.5 What becomes invariant (and why it matters)

With $T \in G_c$, the interval itself is invariant:

(6.6) $\mathbf{s_c^2(e_1,e_2) = s_c^2(T(e_1),T(e_2))}$ (frame-invariant sign classification).

With T_E satisfying Assumption C, compatibility classes are preserved:

- commuting stays commuting, and “no joint POVM exists” stays “no joint POVM exists” after mapping.

Together, (T,T_E) preserve the prerequisites that power AB-fixedness (Section 5), yielding an

invariance theorem: AB-fixedness statements transfer across collapse-frame maps.

6.6 Why this belongs in the merged paper: "small probe change" vs "regime change" as an invariant distinction

PORE's harness insists that a "null probe" must not secretly be Pump/Switch/Couple; otherwise the protocol is not stable and $\hat{\Xi}$-coordinates become observer-sensitive.
Collapse-frame geometry supplies an **observer-transportable** way to formalize the design intent behind "null probe":

- *Small probe change over a few ticks* corresponds to staying in a time-like regime with small $||\Delta\theta||_g$, i.e., $s_{c^2} \geq 0$ or only mildly negative.
- *Regime change* corresponds to a channel-like separation dominated by $||\Delta\theta||_g$, i.e., $s_{c^2} < 0$ with large magnitude—precisely the setting where incompatibility/non-commutation tends to appear in many designs.

The observer paper makes this correlation explicit as a design heuristic with two regimes:

- (H1) **Local-commutation regime:** $\exists\ d_* > 0$ such that $||\Delta\theta||_g \leq d_*$ (plus pointer-preserving/QND structure) implies effects commute.
- (H2) **Incompatibility regime:** $\exists\ d^\dagger > d_*$ such that $||\Delta\theta||_g \geq d^\dagger$ implies selected effects fail to commute (no joint POVM).

Merged interpretation (still operational): choose your PORE Probe pulses so that intended "Probe-only" comparisons stay within the H1 neighborhood (small $||\Delta\theta||_g$), and treat excursions into H2 as candidates for **Switch-like** events (mode changes) or **Gate3 failures** (probe backreaction masquerading as regime shift).

Finally, the counterexamples sharpen the necessity: if your frame map is not isometric (breaks s_{c^2}), or if you move geometry without a matching *-automorphism on effects, then compatibility judgments can be systematically wrong and cross-observer agreement can be spuriously created or destroyed.

7) The core synthesis: an Observer→PORE compiler (main theorem-style section)

Sections 2–6 gave two "hard" objects:

1. a **protocol-first** operational interface PORE, where meaningful coordinates are *compiled* from logged trajectories under a declared protocol P and validated by harness gates ;
2. a **self-referential observer process** that is mathematically well-posed as a unique law on Φ^∞ once measurability/regularity holds (Ionescu–Tulcea), with internal collapse and policy-induced latching .

This section is the integration hinge: we define a protocol-fixed compiler **C_P** that maps observer traces/logs into PORE's effective coordinates $\hat{\Xi}$, and we state a theorem-like proposition whose key feature is an **iff**: *$\hat{\Xi}$ is a coordinate only when the harness passes.*

7.1 Mapping table (observer primitives → PORE primitives)

Observer theory	PORE / PoE protocol

Observer theory	PORE / PoE protocol
tick index k / τ	sampling step Δ and event index n (protocol timebase)
policy θ_k = f_k(trace)	Switch/Probe choice as a declared operator channel (context selection is an explicit act)
instrument M_{θ,φ} and kernel K_k	Probe channel; measured response under Π_probe constraint (probe must be held fixed / declared)
records + compatibility	Couple channel: agreement requires accessible record + commutation/joint measurability after a frame map
between-tick CPTP E_k	Pump-like latent evolution / resource reshaping (declared intervention family or endogenous "plant" evolution)

The mapping is not metaphorical: each row is a constraint about **what must be declared and controlled** if we want "observer talk" to compile into stable operational coordinates.

7.2 The compiler object: what C_P takes in, and what it outputs

PORE's stance is that all downstream claims must be computed from protocol-bounded logs:

(7.1) **P = (B, Δ, h, u)**
(7.2) **z[n] = h(x(to + nΔ))**

For an adaptive observer, the natural "Σ-layer" log is the trace plus declared knobs:

(7.3) **ℓ[n] := (z[n], u[n])**, with u[n] ∈ {Pump, Probe, Switch, Couple}.

A **protocol-fixed compiler** is then a deterministic map on logs (and only logs):

(7.4) **C_P: {ℓ[n]}_{n=0..N} ↦ Ξ̂(L) = (ρ̂(L), γ̂(L), τ̂(L))**, where L is a validated loop under P.

PORE explicitly positions this compiler as the "integration hinge" and insists Ξ̂ is only meaningful when it survives gate checks rather than narrative convenience.

7.3 Harness semantics: what it means for Ξ̂ to be a "coordinate"

We formalize "Ξ̂ is a coordinate" minimally as **repeatable identifiability under a fixed protocol**, which (in PORE) is enforced by at least two gates:

Gate 1: proxy stability (Ξ̂ must be stable under P)

(7.5) **CV_ρ = std({ρ̂(W_k)})/(|mean({ρ̂(W_k)})|+ε0)**
(7.6) **CV_γ = std({γ̂(W_k)})/(|mean({γ̂(W_k)})|+ε0)**
(7.7) **CV_τ = std({τ̂(W_k)})/(|mean({τ̂(W_k)})|+ε0)**

(7.8) **Gate1 pass ⇔ (CV_ρ≤c_ρ) ∧ (CV_γ≤c_γ) ∧ (CV_τ≤c_τ)**

Interpretation: failing Gate 1 means "Ξ̂ is not well-defined under this protocol," i.e., not a usable coordinate system for control or explanation.

Gate 3: probe backreaction sanity (Π_probe must be fixed)

PORE makes "observer inside the system" operational by requiring that a **null probe** be small:

(7.9) **Run a declared null probe δu_Q and require: $||\Delta\hat{\Xi}_Q||_2 \le \theta_Q$ and jump rate does not increase materially.**

And it states the key diagnosis:

(7.10) **If Gate3 fails, Π_probe is not fixed; "measurement" is acting like Pump/ Switch/Couple.**

This is exactly the bridge to the observer model: adaptive observers select contexts; Gate 3 forces us to declare and test whether those context changes are "mere probing" or dynamics-changing interventions.

7.4 Compiler proposition (the paper's hinge)

Proposition 7.1 (Observer→PORE Compilation and Coordinate Validity)

Assume the adaptive observer satisfies the standing conditions for existence/uniqueness:

- Φ finite/countable; Θ standard Borel; measurable policies f_k; between-tick CPTP evolution E_k; normalized instruments M_{θ,φ}.

Then:

1. (**Well-posedness of the source**) There exists a unique probability law P_obs on Φ^∞ induced by the measurable kernels K_k and the tick recursion (Ionescu–Tulcea).
2. (**Protocol-fixed compilation**) For any declared PORE protocol P=(B,Δ,h,u), there exists a deterministic compiler C_P that maps protocol logs ℓ[n] into $\hat{\Xi}(L)$ whenever the loop L is valid under P.
3. (**Coordinate validity iff harness passes**) The compiled $\hat{\Xi}(L)$ is a usable effective coordinate under P **iff** Gate 1 (proxy stability) and Gate 3 (probe backreaction sanity / Π_probe fixedness) both pass.

Proof sketch (operational, minimal).

- *(Existence/uniqueness.)* The observer paper's Theorem 3.4 constructs P_obs uniquely from measurable kernels and the state recursion.
- *(Compilation.)* PORE defines the compiler as a deterministic routine over z[n] (and declared u[n]) under fixed P, producing $\hat{\Xi}(L)$ for validated loops.
- *(Sufficiency.)* If Gate 1 passes, $\hat{\Xi}$ varies slowly enough across windows to serve as a coordinate chart for local modeling/control; if Gate 3 passes, the Probe channel is identifiable as "measurement/interface" rather than hidden control, so comparing $\hat{\Xi}$ across probe conditions is legitimate.
- *(Necessity.)* If Gate 1 fails, the empirical definition of $\hat{\Xi}$ depends on windowing/noise/ drift, so "$\hat{\Xi}$" is not a coordinate under P by PORE's own criterion. If Gate 3 fails, Π_probe is not fixed; the act of "measuring" changes the effective plant, so $\hat{\Xi}$ becomes probe-relative and cannot be treated as protocol-stable without explicitly upgrading the model class (i.e., redefining what Probe means in P).

■

7.5 Why the "iff" matters (one sentence, no metaphysics)

The compiler proposition forces a clean discipline: **observer descriptions only become "portable reality coordinates" when they survive stability and backreaction tests under a declared protocol**; otherwise the correct statement is not "we found a weird ontology," but "$\Xi\hat{}$ is undefined (or probe-relative) under this P."

If you want to continue in the next prompt, the natural next section is **8) Harness gates + Minimal Experiment Protocol**, because Proposition 7.1 becomes practically useful only once we show how to *estimate gains* and *reject regime jumps* in a way that respects latching and the collapse-frame cautions.

8) Harness gates: turning observer talk into testable engineering

Sections 3–7 make "observer" mathematically precise and "protocol" operationally explicit. Section 8 is where the merged framework earns its engineering credibility: we specify **harness gates** that decide whether compiled coordinates are (i) stable enough to deserve the name "coordinate," and (ii) identifiable under a Probe channel that does not secretly alter the plant. In short: *the harness converts interpretive claims into pass/fail diagnostics.*

8.1 Compiled coordinates as protocol artifacts

Given a validated loop L under protocol P, the compiler outputs a loop-local coordinate triple:

(8.1) $\mathbf{\Xi\hat{}(L) = (\rho\hat{}(L), \gamma\hat{}(L), \tau\hat{}(L)).}$

In the PORE vocabulary, $\Xi\hat{}$ is not an ontic state; it is a **reproducible summary of effective loop structure** under the declared interface P=(B,Δ,h,u).

To test reproducibility, we compute $\Xi\hat{}$ over a sequence of sliding windows W_k (all defined under the same P and within the same candidate loop L):

(8.1a) $\mathbf{\Xi\hat{}(W_k) = (\rho\hat{}(W_k), \gamma\hat{}(W_k), \tau\hat{}(W_k)).}$

8.2 Gate 1: proxy stability ($\Xi\hat{}$ must survive windowing)

Gate 1 measures whether the coordinate estimates are stable across windows. For each component $x \in \{\rho\hat{}, \gamma\hat{}, \tau\hat{}\}$, define the coefficient of variation:

(8.2) $\mathbf{CV_\rho = std(\{\rho\hat{}(W_k)\})/(|mean(\{\rho\hat{}(W_k)\})|+\varepsilon_0)}$
(8.3) $\mathbf{CV_\gamma = std(\{\gamma\hat{}(W_k)\})/(|mean(\{\gamma\hat{}(W_k)\})|+\varepsilon_0)}$
(8.4) $\mathbf{CV_\tau = std(\{\tau\hat{}(W_k)\})/(|mean(\{\tau\hat{}(W_k)\})|+\varepsilon_0)}$

Here $\varepsilon_0>0$ is a small numerical stabilizer to avoid division blow-ups near zero. The pass condition is:

(8.5) $\mathbf{Gate1\ pass \Leftrightarrow (CV_\rho \le c_\rho) \wedge (CV_\gamma \le c_\gamma) \wedge (CV_\tau \le c_\tau).}$

Interpretation. If Gate 1 fails, the merged framework does not allow the story "$\Xi\hat{}$ describes the object." Instead it forces one of four protocol actions:

- adjust B (too much leakage / mis-bounded object),
- adjust Δ (aliasing / insufficient sampling resolution),

- adjust h (proxy not aligned with loop invariants),
- split L (you are mixing regimes / latched branches).

This is exactly where the observer-side latching phenomenon matters: a history-dependent policy can create branch-specific futures; Gate 1 failure is the diagnostic signature that your window set {W_k} has crossed an implicit Switch boundary or mixed multiple branches.

8.3 Gate 3: probe backreaction sanity (Π_probe must be fixed)

Gate 3 enforces the key epistemic constraint of the merged paper: **probing is an intervention channel**, and it can invalidate coordinate claims if it changes the effective dynamics.

Define a "null probe" as the smallest declared Probe perturbation δu_Q that is intended to be informational but not dynamical (e.g., a minimal change in measurement context, readout resolution, or sampling micro-protocol).

Let $\hat{\Xi}_0$ be the compiled coordinate under baseline probing, and $\hat{\Xi}_Q$ be the coordinate under the null probe, both evaluated on matched windows (same B,Δ,h and same loop segment):

(8.6) $\mathbf{\Delta\hat{\Xi}_Q := \hat{\Xi}_Q - \hat{\Xi}_0.}$

Gate 3's basic requirement is "smallness":

(8.7) **Gate3 pass** $\Rightarrow \|\Delta\hat{\Xi}_Q\|_2 \le \theta_Q$ **and jump-rate does not increase materially.**

A simple, implementable jump-rate diagnostic is:

(8.8) $\mathbf{J := (1/N)\ \Sigma_{n=1}^{N}\ 1\{\ \|\hat{\Xi}(W_{n}) - \hat{\Xi}(W_{n-1})\|_2 \ge \eta\ \},}$

then require $J_Q \le J_0 + \delta_J$ for a small tolerance δ_J.

If Gate 3 fails, the interpretation is forced (no narrative wiggle room):

(8.9) **Gate3 fail ⇒ "Probe" is not identifiable as Probe; probing is acting like Pump/ Switch/Couple.**

This is the operational analog of the observer-side warning that context changes can latch branches: if a "probe" changes the future kernel family materially, it is not merely reading the system—it is rewriting the loop.

8.4 Why Gate 1 + Gate 3 are the minimal pair for the merged thesis

- Gate 1 enforces **coordinate legitimacy**: $\hat{\Xi}$ must be stable enough to act as a chart.
- Gate 3 enforces **observer-discipline**: measurement must not smuggle in hidden control; otherwise "collapse/agreement" become probe-relative and cannot be compared across observers or protocols without upgrading P.

Together, they implement the Proposition 7.1 "iff": *$\hat{\Xi}$ exists as a portable, protocol-fixed object only inside the region where proxies are stable and probe backreaction is controlled.*

9) Minimal experiment protocol (MEP) for observers (gain estimation)

Sections 7–8 established that $\hat{\Xi}$ is only meaningful when it is **protocol-stable** (Gate 1) and

probe-identifiable (Gate 3). Section 9 adds the smallest practical routine that turns the merged framework into a *working lab method*: estimate how each operator channel {Pump, Probe, Switch, Couple} moves the compiled coordinates, and use that to **separate "measurement" from "regime change"** with explicit rejection rules.

9.1 Local linear response model in Ξ-space

Fix a protocol P=(B,Δ,h,u) and a validated loop segment where Gate 1 and Gate 3 pass. Choose a local operating point $\bar{\Xi}$ (e.g., running mean over a calibration window). Define deviations:

(9.0) $\delta\Xi_t := \hat{\Xi}_t - \bar{\Xi}$, where $\hat{\Xi}_t$ abbreviates $\hat{\Xi}$ evaluated on the t-th window/step.

Let δu_t be the declared intervention at step t in the 4-channel operator basis (Pump, Probe, Switch, Couple). The local discrete response model is:

(9.1) $$\delta\Xi_{t+1} = \tilde{A}\,\delta\Xi_t + \hat{G}\,\delta u_t + \xi_t.$$

where:

- $\tilde{A} \in \mathbb{R}^{3\times 3}$ is the local drift/return map in Ξ-space,
- $\hat{G} \in \mathbb{R}^{3\times 4}$ is the channel gain matrix,
- ξ_t is residual noise + unmodeled dynamics (including observer-policy microstructure not captured by $\hat{\Xi}$).

This is not claimed as globally valid—only as a **local instrument** for protocol diagnosis and operator disentanglement.

9.2 One-channel pulses and gain matrix estimation

Write δu_t in the canonical operator basis:

(9.2a) $$\delta u_t = (\delta u_{P,t}, \delta u_{Q,t}, \delta u_{Sw,t}, \delta u_{C,t})^{T}.$$

Then the gain matrix is column-partitioned as:

(9.2) $$\hat{G} = [\hat{g}_P\ \hat{g}_Q\ \hat{g}_{Sw}\ \hat{g}_C] \in \mathbb{R}^{3\times 4},$$

where $\hat{g}_P$ is the Ξ-response direction to a unit Pump pulse, $\hat{g}_Q$ to a unit Probe pulse, etc.

MEP pulse design (minimal, reproducible).

1. Choose a baseline segment with stable $\hat{\Xi}$ (Gate 1 pass).
2. Apply **one-channel pulses**: vary only one component of δu_t at a time while holding the others at 0 (or at declared baseline values).
3. Use matched windows: keep Δ, h, and boundary B fixed, and keep pulse magnitudes small enough to stay in the same local regime unless you are *testing* for regime change.

A simplest estimator (when $\tilde{A}$ is weak or when you difference out the drift) is:

(9.3) $$\Delta\Xi_t := \hat{\Xi}_{t+1} - \hat{\Xi}_t \approx \hat{G}\,\delta u_t + \varepsilon_t,$$

estimated by least squares over pulse trials. More generally, estimate $\tilde{A}$ and $\hat{G}$ jointly via regression on $(\delta\Xi_t, \delta u_t)$.

9.3 Probe vs Switch: empirical separation by rejection rules

The merged paper needs a practical way to distinguish:

- **Probe**: "I queried differently, but did not change the effective plant," versus
- **Switch**: "I (or the system) entered a different regime; comparisons across the boundary require re-compilation."

This is where MEP + harness gates become a decision procedure.

9.3.1 The Probe signature (what Probe is allowed to do)

Under a *valid* Probe channel, a null (small) Probe pulse should produce:

- small $\Delta\hat{\Xi}$ (Gate 3), and
- no persistent change in drift ($\tilde{A}$) beyond noise.

Operationally, we require both:

(9.4) $\|\hat{g}_Q\|_2$ **is small relative to** $\|\hat{g}_P\|_2$ **and** $\|\hat{g}_C\|_2$ (Probe is "mostly readout," not "resource reshaping"), and

(9.5) $\tilde{A}_{pre} \approx \tilde{A}_{post}$ (no systematic drift change after Probe pulse).

These are empirical statements: if they fail, Probe is not behaving as Probe under this protocol.

9.3.2 Switch/KL-like event signature (what counts as regime change)

A regime change is declared when at least one of the following rejection rules triggers:

Rule S1 (jump magnitude):

(9.6) $\|\hat{\Xi}_{t+1} - \hat{\Xi}_t\|_2 \geq \eta_{Sw}$.

Rule S2 (model mismatch surge):

Let r_t be the one-step prediction residual:

(9.7) $r_t := \delta\Xi_{t+1} - \tilde{A}\,\delta\Xi_t - \hat{G}\,\delta u_t$.

Declare Switch if:

(9.8) $\|r_t\|_2 \geq r_*$ *for M consecutive steps.**

Rule S3 (drift matrix change):

Estimate $\tilde{A}$ on windows before and after the candidate event, $\tilde{A}_{pre}$ and $\tilde{A}_{post}$, and declare Switch if:

(9.9) $\|\tilde{A}_{post} - \tilde{A}_{pre}\|_F \geq \alpha^*$.

Rule S4 (Probe identifiability failure):

If the event occurred under a declared Probe pulse ($\delta u_{Q,t} \neq 0$) but Gate 3 fails, then treat the event as Switch-like (or as "Probe→intervention misclassification" requiring protocol upgrade):

(9.10) **Gate3 fail under Probe ⇒ reclassify as Switch/Couple/Pump candidate.**

These rules instantiate the conceptual point from Section 6: “small probe change” should stay in a local neighborhood (small channel distance / no algebraic incompatibility), whereas regime change corresponds to leaving that neighborhood, producing nonlocal shifts in $\hat{\Xi}$ and/or the local linear model.

9.4 What the MEP buys the merged thesis (one paragraph)

MEP converts the observer model’s core tension—**measurement choices are part of dynamics**—into a measurable operator decomposition: $\hat{G}$ explicitly quantifies how much of “what happened” is attributable to Pump vs Probe vs Switch vs Couple under a fixed protocol. In particular, it gives a concrete test for the central discipline claim: if Probe pulses systematically yield Switch-like signatures (jump rules, residual surges, drift changes), then the protocol has **ontology drift masquerading as observation**, and the correct response is to re-declare P (adjust B/Δ/h or split loops), not to argue about interpretation.

10) Discussion: philosophical core, non-claims, and failure modes

This merged framework is deliberately “post-ontological”: it does not try to win a metaphysical contest about what quantum states *are*. Instead, it sets up a disciplined pipeline in which (i) observer dynamics are made mathematically well-posed and (ii) any claim of “stable effective reality” must survive protocol-fixed compilation and harness gates. The philosophical stance is therefore not an opinion layered on top of physics, but a **constraint system** that prevents silent shifts in meaning.

10.1 Philosophical core (what the merge commits to)

10.1.1 Protocol realism (weak, operational)

The basic commitment is:

(10.1) **Meaningful claims are protocol-indexed: Claim(P) rather than Claim(Reality).**

A “thing” is treated as an **effective loop-object under P**—a recurrent, controllable, measurable regularity—so changing P changes the object unless equivalence is demonstrated.

10.1.2 Collapse is internal (filtration-fixedness + latching)

The observer formalism yields a precise statement:

- collapse = **internal conditional certainty**:
    - realized past events are delta-certain under conditioning on the observer filtration F_k.
- branching irreversibility = **policy-induced latching**:
    - different realized histories yield different future contexts and kernels.

So “collapse” is not imported as a new dynamical axiom; it is a consequence of a well-defined adaptive stochastic process plus the observer’s own information structure.

10.1.3 Objectivity is engineered (records + compatibility + redundancy)

Cross-observer agreement is not assumed; it is obtained when:

- there is a **frame map** aligning propositions/effects,
- **compatibility** holds (commutation/joint measurability),
- and a **record** is accessible (so conditioning is meaningful).

"Objectivity" in the many-observer sense emerges when records are **redundantly encoded** (SBS/quantum Darwinism style), so different observers can read different fragments and still converge with high probability.

10.2 Non-claims (explicit scope limits)

To prevent category errors, we state what is **not** being claimed.

1. **Not an interpretation of quantum mechanics.**
   No new postulate is added to the Born rule/instrument formalism; the framework reorganizes standard ingredients around adaptive observers and protocol discipline.

2. **Not a collapse mechanism.**
   We do not claim a physical "trigger" for collapse. "Collapse" is used strictly in the internal sense (filtration-fixedness) plus policy-latching as an operational branching source.

3. **Not an a priori guarantee of objectivity.**
   Objectivity is conditional: if compatibility, records, and redundancy fail, there is no obligation for observers to agree.

4. **Not a universal TOE claim.**
   PORE's universality is methodological: it offers a portable protocol grammar and falsifiability harness; it does not assert a unique underlying decomposition of all phenomena.

10.3 Failure modes as epistemic boundaries (where statements become undefined)

A central value of the merged framework is that it draws **hard edges**: certain failures do not merely increase uncertainty; they make the question ill-posed under the protocol.

10.3.1 Observer-model boundary: non-measurable policy ⇒ no global law

The observer's existence/uniqueness result relies on measurability of the adaptive kernel family. If the policy f_k is not measurable (or if the induced kernels fail measurability), the Ionescu–Tulcea construction can fail, meaning:

(10.2) **Non-measurable f_k ⇒ global trace law P_obs on Φ^∞ may not exist (ill-posed observer).**

Operational consequence: "what the observer will see in the long run" is not merely unknown—it is not mathematically defined in the model class.

10.3.2 PORE boundary: unstable proxies ⇒ Ξ is not a coordinate

PORE refuses to let unstable summaries masquerade as states. If Gate 1 fails (proxy instability across windows), then:

(10.3) **Gate1 fail ⇒ $\hat{\Xi}$ is not a protocol-stable coordinate under P (coordinate undefined).**

Operational consequence: you must revise B/Δ/h or split the loop (often corresponding to hidden latching/Switch events) rather than arguing about "true values."

10.3.3 Agreement boundary: incompatibility or missing records ⇒ no agreement theorem

AB-fixedness and SBS-style objectivity require compatibility and accessible records. If effects do not commute (no joint measurability), or if B cannot access A's record (no shared or redundant register), then:

(10.4) **(No compatibility) ∨ (No record access) ⇒ cross-observer agreement is not derivable and may fail.**

Operational consequence: disagreement is not paradoxical; it is a diagnosis of the interface—either the propositions are not aligned, or they are not jointly testable, or the record is not actually shared.

10.4 The "one-sentence" merged worldview (kept operational)

Within this framework:

- "Reality for an observer" is a **well-posed trace law** plus **internal conditional certainty**,
- "Shared reality" is **agreement enabled by compatibility + records**,
- "Objective reality" is **redundant records** that many observers can read with exponentially improving consensus,
- and "scientific discipline" is the insistence that any effective coordinate ($\hat{\Xi}$) exists only where the harness gates pass under a declared protocol.

Reference

The Post-Ontological Reality Engine (PORE)
https://osf.io/nq9h4/files/osfstorage/699b33b78ef8cded146cbd5c

Self-Referential Observers in Quantum Dynamics: A Formal Theory of Internal Collapse and Cross- Observer Agreement
https://aixiv.science/pdf/aixiv.251123.000001

Appendix A) One-page Compiler Card (Protocol + Observer Bundle + Gates)

This appendix is a portable artifact: a single-page checklist that lets a reader (or lab / agent engineer) **instantiate** the merged framework without re-reading the paper. It is written as a "do-this-then-that" compilation card, with only the minimum symbols needed to reproduce the Observer→PORE pipeline.

A.0 Inputs and outputs (what you must declare)

Inputs (declared):

- Protocol: (A.0.1) **$P = (B, \Delta, h, u)$**, $u \in \{\text{Pump, Probe, Switch, Couple}\}$.
- Observer bundle: $(\Phi, \Theta, \{f_k\}, \{M_{\theta,\varphi}\}, \{E_k\})$.

Outputs (compiled, only if gates pass):

- (A.0.2) **$\hat{\Xi}(L) = (\hat{\rho}(L), \hat{\gamma}(L), \hat{\tau}(L))$**
- (A.0.3) **$\hat{G} = [\hat{g}_P\ \hat{g}_Q\ \hat{g}_{Sw}\ \hat{g}_C] \in \mathbb{R}^{3\times4}$** (optional, from MEP).

A.1 Protocol card (PORE / PoE)

A.1.1 Boundary B (what counts as the loop-object)

Declare:

- Inside variables (what "belongs to the object" under this protocol),
- Outside drivers (treated as exogenous),
- What records are considered accessible (for agreement tests).

A.1.2 Timebase Δ (what one "tick" means)

Declare:

- sampling cadence Δ,
- window length $|W|$ and step size (for sliding-window estimates).

A.1.3 Observation map h (what gets logged)

Declare:

- (A.1.1) **$z[n] = h(x(t_o + n\Delta))$**
- any preprocessing/embedding of $z[n]$ allowed.

A.1.4 Operator channels u (what you are allowed to do)

Declare for each tick n:

- $u[n] \in \{\text{Pump, Probe, Switch, Couple}\}$
- magnitude convention for pulses δu.

A.2 Observer bundle card (self-referential observer)

A.2.1 Trace space and filtration

Declare:

- outcomes Φ, contexts Θ,
- (A.2.1) $\mathbf{F_k := \sigma(\varphi_1,\ldots,\varphi_k)}$.

A.2.2 Adaptive policy (must be measurable)

Declare:

- (A.2.2) $\mathbf{f_k: \Phi^{k-1} \to \Theta}$ measurable,
- (A.2.3) $\mathbf{\theta_k := f_k(\varphi_{1:k-1})}$.

A.2.3 Instrument family and effects

Declare:

- instrument maps $M_{\theta,\varphi}$ (CP, trace–nonincreasing),
- normalization $\Sigma_\varphi M_{\theta,\varphi}$ CPTP,
- effects $E_{\theta,\varphi} := M^*_{\theta,\varphi}(I)$.

A.2.4 Tick recursion

Declare:

- between-tick evolution (A.2.4) $\mathbf{\rho_{k^-} := E_k(\rho_{k-1})}$ (CPTP),
- kernel (A.2.5) $\mathbf{K_k(\varphi_{1:k-1},\varphi) := Tr[M_{f_k(\varphi_{1:k-1}),\varphi}(\rho_{k^-})]}$,
- update (A.2.6) $\mathbf{\rho_k := M_{\theta_k,\varphi_k}(\rho_{k^-}) / Tr[M_{\theta_k,\varphi_k}(\rho_{k^-})]}$.

A.3 Compilation steps (Observer→PORE)

Step A: Validate “loop existence” under P (Gate0)

Check that the candidate object supports:

- recurrence / return structure,
- bounded leakage (closure under B),
- enough stationarity for windowed estimation.

If Gate0 fails → stop: $\Xi\hat{}$ undefined under this P.

Step B: Compile $\Xi\hat{}$ on sliding windows W_k

Compute:

- (A.3.1) $\mathbf{\Xi\hat{}(W_k) = (\rho\hat{}(W_k), \gamma\hat{}(W_k), \tau\hat{}(W_k))}$

according to the paper's chosen compiler (from logs z[n], u[n]) under fixed P.

Step C: Gate 1 (proxy stability)

Compute coefficients of variation:

- (A.3.2) **$CV_x = std(\{x(W_k)\})/(|mean(\{x(W_k)\})|+\varepsilon_0)$**, $x \in \{\rho\hat{}, \gamma\hat{}, \tau\hat{}\}$

Pass rule:

- (A.3.3) **Gate1 pass $\Leftrightarrow$ $CV_\rho \le c_\rho \wedge CV_\gamma \le c_\gamma \wedge CV_\tau \le c_\tau$.**

If Gate1 fails → action: revise B/Δ/h or split L (likely mixed regimes / latching).

Step D: Gate 3 (probe backreaction sanity)

Run a declared null probe δu_Q and compare compiled coordinates:

- (A.3.4) **$\Delta\Xi\hat{}_Q := \Xi\hat{}_Q - \Xi\hat{}_0$**
  Require:

- (A.3.5) **$||\Delta\Xi\hat{}_Q||_2 \le \theta_Q$** and jump-rate not increased (optional J-test).

If Gate3 fails → diagnosis: Probe is not identifiable; it is acting like Pump/Switch/Couple under this P.

Step E (optional): Minimal Experiment Protocol (MEP) for gains

Fit local linear response:

- (A.3.6) **$\delta\Xi_\{t+1\} = \tilde{A}\,\delta\Xi_t + \hat{G}\,\delta u_t + \xi_t$**
  with:

- (A.3.7) **$\hat{G} = [\hat{g}_P\ \hat{g}_Q\ \hat{g}_{Sw}\ \hat{g}_C] \in \mathbb{R}^{\{3\times 4\}}$**.

Use rejection rules to label regime changes (Switch-like events):

- large jumps in $\Xi\hat{}$, residual surges, drift changes, or Gate3 failures under Probe.

A.4 Agreement / objectivity add-on (cross-observer checklist)

When comparing observers A and B, require:

1. **Frame map** (contexts/outcomes/effects aligned),
2. **Compatibility** (commutation or joint measurability),
3. **Accessible record** (record is measurable in B's filtration).

For many observers, check redundancy (SBS-style):

- (A.4.1) **$\rho_\{S\ E_1 \ldots E_m\} = \Sigma_i\, p_i\, |\psi_i\rangle\langle\psi_i|_S \otimes (\otimes\{j=1\}^m\, \rho_\{E_j\}^{(i)})$**
  and fragment distinguishability so consensus error decays with $|J|$.

A.5 Minimal “stop / proceed” rule (the discipline clause)

- **Proceed** with $\hat{\Xi}$-based modeling/control only if: Gate0 pass ∧ Gate1 pass ∧ Gate3 pass.
- Otherwise the correct statement is: **$\hat{\Xi}$ is undefined or probe-relative under this P**, and the correct move is to revise protocol (B,Δ,h,u) or split the system into multiple loops.

Appendix B) Minimal simulation loop for the observer process

This appendix gives a **minimal, reproducible simulation loop** for the self-referential observer defined in Sections 3–4: an adaptive policy chooses a context, an instrument induces a Born-rule kernel, an outcome is sampled, and the (selective) post-measurement state is updated.

B.1 Objects you must specify

Spaces

- Outcome alphabet: Φ (finite or countable).
- Context space: Θ (in simulation, any indexable set is fine).

State

- Hilbert space dimension d and initial density matrix $\rho_0 \in \mathbb{C}^{d\times d}$, $\rho_0 \succeq 0$, $\mathrm{Tr}(\rho_0)=1$.

Between-tick evolution

- CPTP maps $\{E_k\}$ (k=1..K), applied before each measurement tick.

Instrument family

- For each context $\theta \in \Theta$ and outcome $\varphi \in \Phi$, a CP trace–nonincreasing map $M_{\theta,\varphi}$.
- Normalization: $\Sigma_{\varphi\in\Phi} M_{\theta,\varphi}$ is CPTP for every θ.

A convenient Kraus form:

(B.1) $\mathbf{M_{\theta,\varphi}(\rho) := \Sigma_{a=1}^{A(\theta,\varphi)} K_{\theta,\varphi,a}\, \rho\, K^{\dagger}_{\theta,\varphi,a}}$
(B.2) $\mathbf{\Sigma_{\varphi\in\Phi} \Sigma_a K^{\dagger}\{\theta,\varphi,a\}\, K\{\theta,\varphi,a\} = I}$ (instrument normalization)

Policy

- Adaptive measurable policy family $\{f_k\}$:

(B.3) $\mathbf{\theta_k := f_k(\varphi_{1:k-1})}$

In code, this is just a deterministic function from the past trace to a context.

B.2 Minimal tick loop (mathematical form)

At tick k:

(B.4) **ρ_{k^-} := E_k(ρ_{k−1})**
(B.5) **p_k(φ) := Tr[M_{θ_k,φ}(ρ_{k^-})]** (Born-rule kernel)
(B.6) **φ_k ~ Categorical({p_k(φ)}_{φ∈Φ})**
(B.7) **ρ_k := M_{θ_k,φ_k}(ρ_{k^-}) / p_k(φ_k)**

This generates a sample path φ_{1:K} ∈ Φ^K together with updated states {ρ_k}.

B.3 Minimal pseudocode (language-agnostic)

```
INPUT: K, rho0, policy f_k, CPTP evolutions E_k, instruments M(theta,phi)
OUTPUT: trace phi[1..K], contexts theta[1..K], states rho[0..K]

rho = rho0
phi_trace = []

for k in 1..K:
    # (1) between-tick evolution
    rho_minus = E_k(rho)

    # (2) choose context from past outcomes
    theta = f_k(phi_trace)

    # (3) compute Born probabilities over outcomes
    for each phi in Φ:
        unnorm = M(theta, phi, rho_minus)          # CP map output (matrix)
        p[phi] = trace(unnorm)

    normalize p so sum_phi p[phi] = 1

    # (4) sample outcome
    phi_k = sample_categorical(p)

    # (5) selective update
    rho = M(theta, phi_k, rho_minus) / p[phi_k]

    append phi_k to phi_trace
    store theta, rho
```

B.4 Minimal Python-like skeleton (placeholders for your Kraus ops)

```
import numpy as np

def tr(A):
    return np.trace(A).real

def simulate_observer(K, rho0, Phi, policy_f, E_map, M_map, rng=None):
    """
    K: number of ticks
    rho0: density matrix (dxd)
    Phi: list of outcomes
    policy_f: function f_k(k, phi_trace) -> theta
    E_map: function E_k(k, rho) -> rho_minus (CPTP)
    M_map: function M(theta, phi, rho_minus) -> unnormalized rho (CP, trace-nonincreasing)
    """
    rng = np.random.default_rng() if rng is None else rng

    rho = rho0.copy()
    phi_trace = []
    theta_trace = []
    rho_trace = [rho.copy()]

    for k in range(1, K + 1):
```

```
        rho_minus = E_map(k, rho)

        theta = policy_f(k, phi_trace)

        probs = []
        unnorms = []
        for phi in Phi:
            un = M_map(theta, phi, rho_minus)
            unnorms.append(un)
            probs.append(max(tr(un), 0.0))

        Z = sum(probs)
        if Z <= 0:
            raise ValueError("All outcome probabilities are zero; instrument/state mismatch.")
        probs = [p / Z for p in probs]

        idx = rng.choice(len(Phi), p=probs)
        phi_k = Phi[idx]

        rho = unnorms[idx] / probs[idx]  # selective update
        rho = (rho + rho.conj().T) / 2   # optional: symmetrize for numerical stability
        rho = rho / tr(rho)             # renormalize trace

        phi_trace.append(phi_k)
        theta_trace.append(theta)
        rho_trace.append(rho.copy())

    return {"phi": phi_trace, "theta": theta_trace, "rho": rho_trace}
```

B.5 Sanity checks (cheap but important)

At each tick, assert:

- **Probability normalization:** $\Sigma_{\varphi\in\Phi} p_k(\varphi) \approx 1$.
- **State validity:** $\rho_k \succeq 0$ (numerically: eigenvalues $\geq -\text{tol}$) and $\mathrm{Tr}(\rho_k)=1$.
- **Instrument normalization:** $\Sigma_{\varphi,a} K^\dagger K = I$ (verify once per θ in setup).

These checks correspond to the paper's "well-posedness" requirements that make the infinite-trace law exist (in the mathematical model) and make finite simulations stable (in practice).

Optional hook to PORE logs (if you want the sim to feed the compiler)

If you want this observer simulator to produce **PORE-ready logs**, define a protocol observation map h and record:

(B.8) **$z[n] = h(\rho_n)$** (e.g., selected expectation values $\mathrm{Tr}(O_i \rho_n)$)
(B.9) **$\ell[n] = (z[n], u[n])$** with $u[n] \in$ {Pump, Probe, Switch, Couple}

This makes the same run usable both as an observer trace generator and as input to the Observer→PORE compilation and gate tests.

Appendix C) Worked toy example: two observers, commuting vs non-commuting effects, SBS vs no-SBS

This appendix gives a concrete "sandbox" where every concept in Sections 3–6 can be pointed to explicitly: **(i)** AB-fixedness (frame map + compatibility + record access), **(ii)** why non-

commuting effects block joint truth assignments, and **(iii)** how **redundant records** (SBS/"quantum Darwinism" style) upgrade pairwise agreement into many-observer objectivity.

C.1 Setup: a qubit system S and two environment fragments E_A, E_B

Let S be a qubit with pointer basis {|0⟩,|1⟩}. Two observers A and B can each read a different environment fragment: A reads E_A, B reads E_B.

We consider two families of joint states.

C.1.1 Ideal SBS-like (perfect redundant records)

A clean "broadcast" pure-state prototype is:

(C.1) $|\Psi_SBS\rangle = \sqrt{p}\,|0\rangle_S\,|0\rangle_{E_A}\,|0\rangle_{E_B} + \sqrt{(1-p)}\,|1\rangle_S\,|1\rangle_{E_A}\,|1\rangle_{E_B}.$

The corresponding mixed SBS form (the one used in the main text) is:

(C.2) $\rho_{S\,E_A\,E_B} = \Sigma_{i\in\{0,1\}}\, p_i\, |i\rangle\langle i|_S \otimes (\rho_{E_A}^{(i)} \otimes \rho_{E_B}^{(i)}),$

with p_0=p, p_1=1−p, and (in the ideal case) the fragment encodings are perfectly distinguishable.

C.1.2 No-SBS (no local record in at least one fragment)

A minimal "no local record for A" counterexample is:

(C.3) $\rho_{S\,E_A\,E_B} = \Sigma_{i\in\{0,1\}}\, p_i\, |i\rangle\langle i|_S \otimes (\rho_{E_A} \otimes \rho_{E_B}^{(i)}),$

i.e., $\rho_{E_A}^{(0)} = \rho_{E_A}^{(1)} = \rho_{E_A}$. Then E_A contains *no information* about i, even though E_B might.

C.2 Commuting case: AB-fixedness becomes "obvious" under SBS

C.2.1 Measurements (effects) that commute

Let A measure E_A in the {|0⟩,|1⟩} basis and B measure E_B in the same basis. Their effects can be written as:

(C.4) $F^A_a = |a\rangle\langle a|_{E_A} \otimes I_{E_B},\ a\in\{0,1\}$
(C.5) $F^B_b = I_{E_A} \otimes |b\rangle\langle b|_{E_B},\ b\in\{0,1\}$

Then:

(C.6) $[F^A_a, F^B_b] = 0$ for all a,b (they act on different subsystems).

So compatibility holds in the strongest way: commuting effects.

C.2.2 Record accessibility and AB-fixedness

Under the SBS state (C.1)/(C.2), A's outcome a is literally a record of i, and B's outcome b is the same record of i. In the ideal case:

(C.7) **P(a=b)=1** and **P(i=a)=1**, hence both observers become delta-certain about the same pointer label after reading their fragment.

This is the toy realization of the AB-fixedness checklist from Section 5:

1. **Frame map:** both observers' "0/1" labels refer to the same pointer label i.
2. **Compatibility:** effects commute (C.6).
3. **Accessible record:** each observer has a local record in their filtration (they can read it).

C.2.3 Imperfect records: disagreement probability and redundancy

Now let each fragment encode i with non-orthogonal pure states $|e_i\rangle$, $|f_i\rangle$ (still product across fragments):

(C.8) $|\Psi\rangle = \sqrt{p}\,|0\rangle|e_0\rangle|f_0\rangle + \sqrt{(1-p)}\,|1\rangle|e_1\rangle|f_1\rangle$,

with overlaps $\alpha := |\langle e_0|e_1\rangle|$ and $\beta := |\langle f_0|f_1\rangle|$.

Each observer can misidentify i. The key qualitative fact is: if you have **m independent readable fragments**, the optimal discrimination error typically decays exponentially with m under standard distinguishability bounds (Chernoff/trace-distance families), matching the "redundancy improves consensus" claim in Section 5.

C.3 Non-commuting case: why compatibility is not optional

To see the obstruction cleanly, make A and B measure **incompatible propositions on the same qubit S**.

Let A measure σ_z (Z-basis) on S, and B measure σ_x (X-basis) on S. Define effects:

(C.9) $E^A_0 = |0\rangle\langle 0|$, $E^A_1 = |1\rangle\langle 1|$
(C.10) $E^B_+ = |+\rangle\langle +|$, $E^B_- = |-\rangle\langle -|$

with $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Then:

(C.11) $[E^A_0, E^B_+] \neq 0$ (and similarly for others).

C.3.1 What fails, concretely

If the system is in $|0\rangle$, then A's outcome is delta-certain:

(C.12) $P_A(0 \mid Z, |0\rangle)=1$.

But B's X-measurement is not:

(C.13) $P_B(+ \mid X, |0\rangle)=1/2$, $P_B(- \mid X, |0\rangle)=1/2$.

So you cannot demand a single joint assignment that makes both "A saw Z=0" and "B will see X=+ with certainty" true simultaneously. This is exactly why the AB-fixedness condition includes **compatibility/joint measurability**: without it, "agreement" is not a theorem you're allowed to ask for.

C.3.2 Subtle but important: record-sharing is not the same as compatibility

Even if B later learns A's record classically ("A recorded 0"), B can agree **about A's statement**. That is record accessibility. But it does not change the fact that X and Z are incompatible propositions about S. So:

- record access enables *agreement on reports*,
- compatibility enables *joint probabilistic truth assignments about the same system under the same event structure*.

C.4 SBS vs no-SBS: why redundancy is the objectivity switch

C.4.1 SBS: both have local records ⇒ robust agreement

Under (C.2), each fragment marginal depends on i:

(C.14) $\rho_{E_A}^{(0)} \neq \rho_{E_A}^{(1)}$ and $\rho_{E_B}^{(0)} \neq \rho_{E_B}^{(1)}$.

So both observers can, in principle, extract i locally. With multiple fragments, many observers can read different subsets and still converge with high probability. This is the many-observer "objectivity" mechanism described in Section 5.

C.4.2 No-SBS: one observer lacks a record ⇒ agreement is not guaranteed

Under (C.3), the marginal on E_A carries no information about i:

(C.15) $\rho_{S\,E_A} = \Sigma_i\, p_i\, |i\rangle\langle i| \otimes \rho_{E_A} = (p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|) \otimes \rho_{E_A}$.

So any measurement on E_A yields statistics independent of i. Observer A cannot form a local record, hence AB-fixedness fails at condition (3) (accessible record), and "objectivity" fails because redundancy is missing on A's side.

C.5 How this toy example interfaces with PORE (one operational translation)

- In PORE terms, **SBS corresponds to a strong Couple channel**: the system's label i is redundantly bound into multiple accessible registers (environment fragments), stabilizing cross-observer agreement.
- The commuting vs non-commuting contrast is the sharpest illustration of why "small probe change vs regime change" cannot be decided by geometry alone: it is ultimately about whether the transported effects preserve commutators (Section 6's warning).
- Finally, this sandbox shows why PORE's Gate 3 matters: if B's "probe" choice moves from a commuting neighborhood (compatible readout of a shared record) into an incompatible one (non-commuting effects), then what looks like "just measurement" is actually a **Switch-like boundary** for comparability.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.

M

T

G

Text-to-speech function is limited to 200 characters